

Data Analysis

Final assignment

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Table of contents

1. Introduction
 2. Data preprocessing
 3. Discussion
 4. Probability Analysis
 5. Law of large numbers
 6. Central limit theorem
 7. Actual vs. Predicted sales
 8. PCA
 9. t-SNE
 10. UMAP
 11. Chi-square test
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Introduction

We analyzed **Rossmann Store Sales** found on Kaggle: [link](#)

The dataset has:

1. *1115 stores*
2. *750-980 rows* for each store.
3. Each row consists of *Sales, Promotions, Customers*, etc.

Store	DayOfWeek	Date	Sales	Customers	Open	Promo	StateHoliday	SchoolHoliday
1	5	31/07/2015	5263	555	1	1	0	1
2	5	31/07/2015	6064	625	1	1	0	1
3	5	31/07/2015	8314	821	1	1	0	1

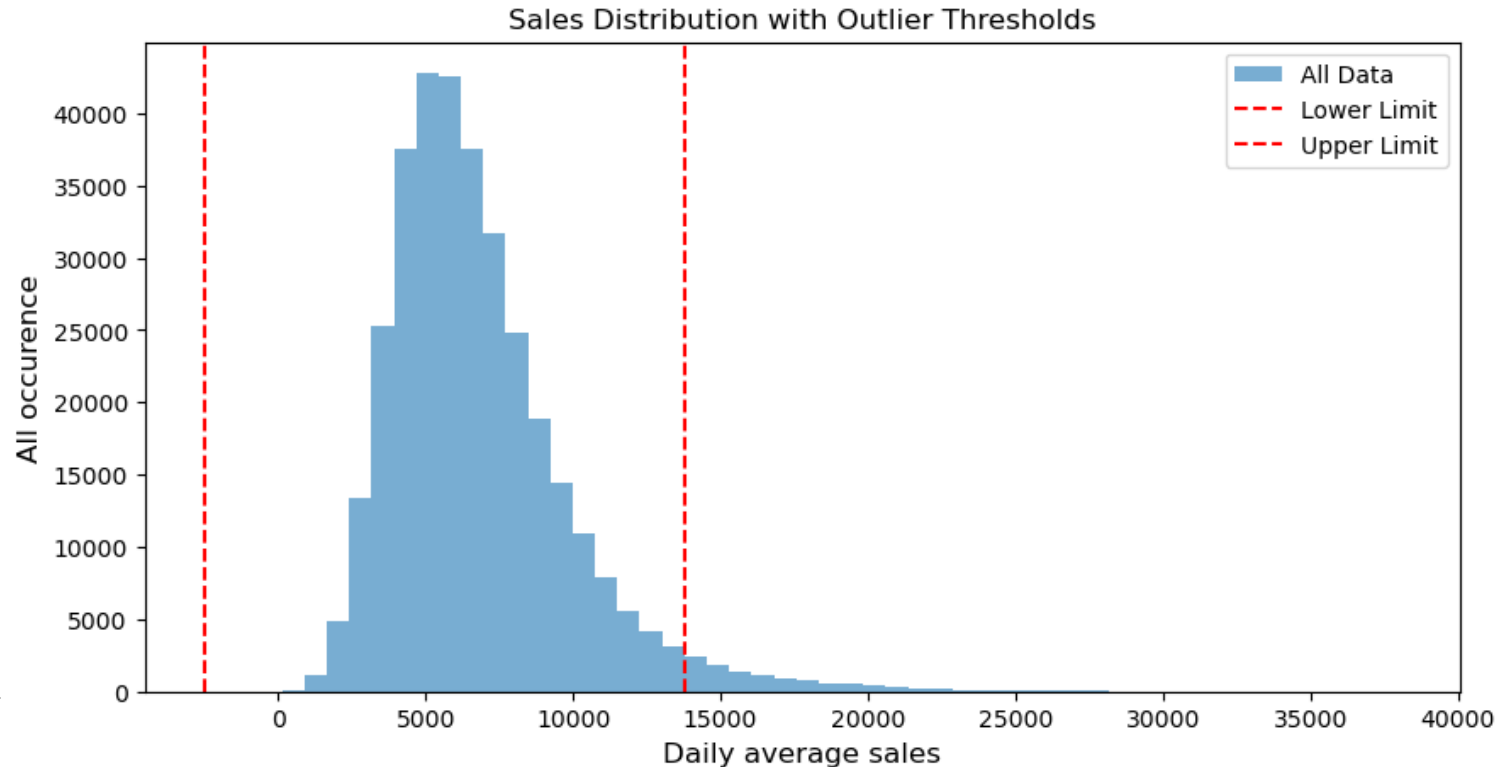
Dataset shape: 1,017,209 x 9

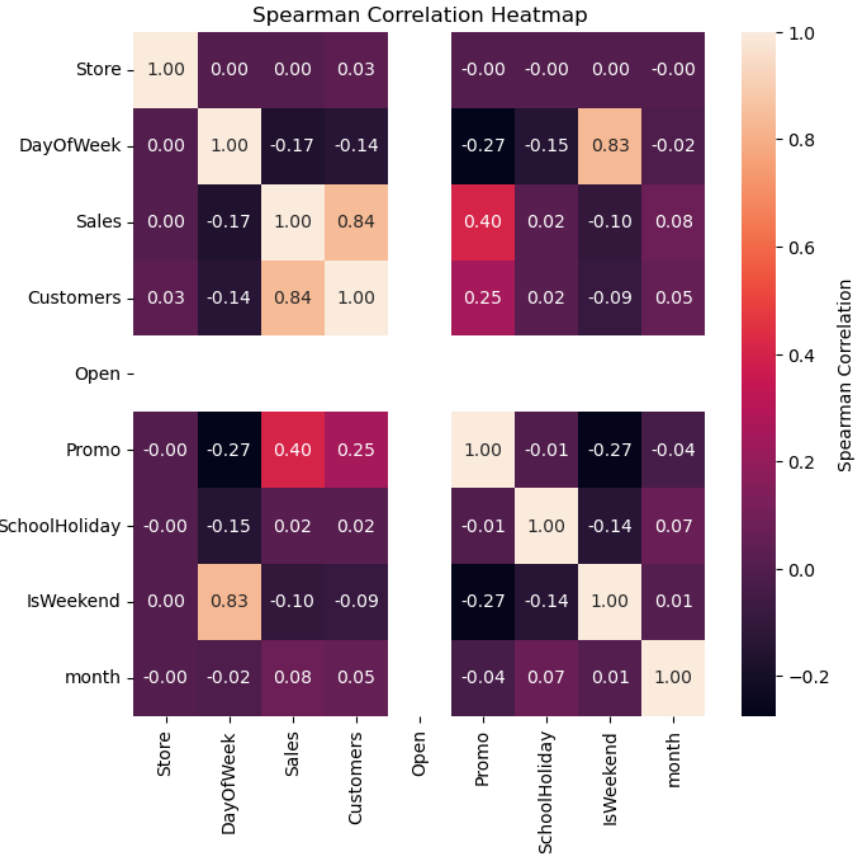
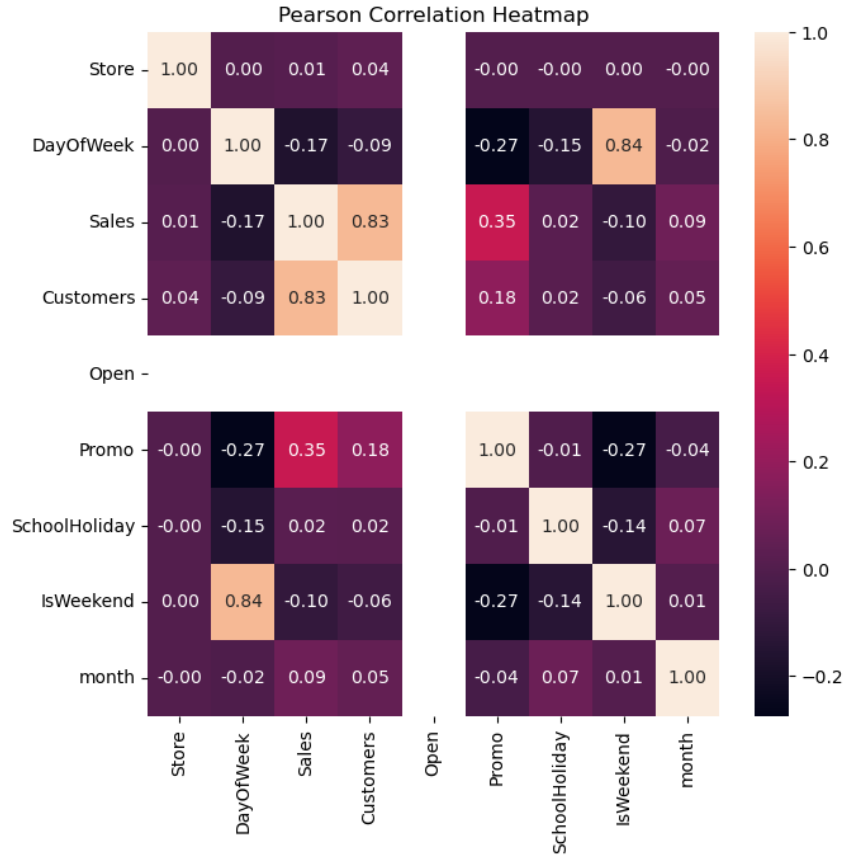
Data preprocessing

1. Extracted: 406,974 rows
2. Null values: 0 cells
3. Duplicates: 0 rows
4. No sale days: 69,050 (removed)
5. Outliers: 11,167 (not removed)
6. Features engineered: "IsWeekend"

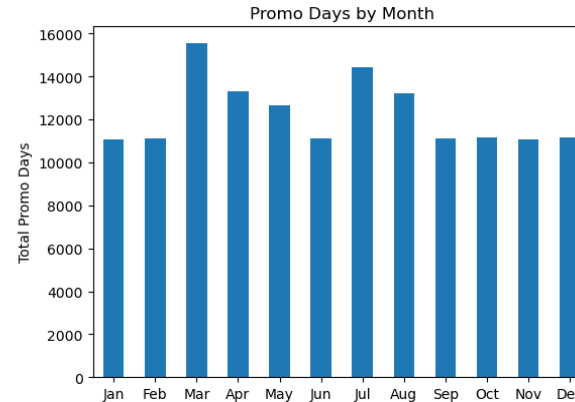
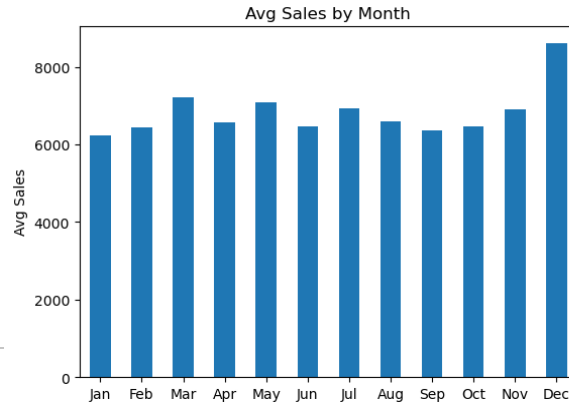
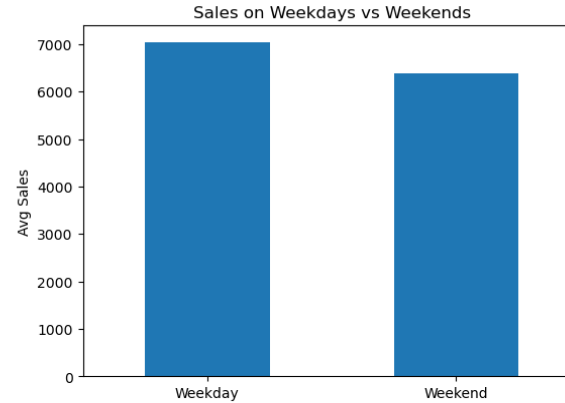
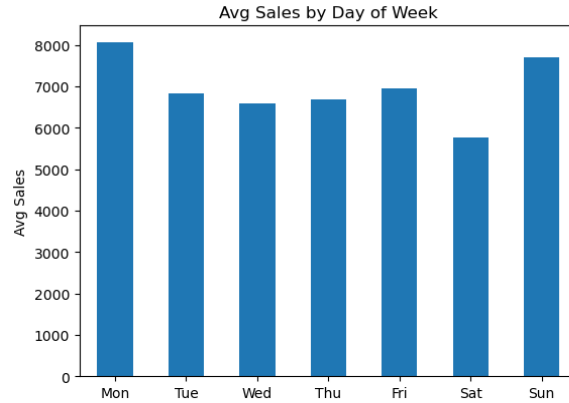
Dataset shape: 337,924 x 10

Outliers





Daily/Periodic Pattern Analysis of all stores



Probability analysis

$P(\text{HighSales} \mid \text{Promo}=1): 18.03 \%$

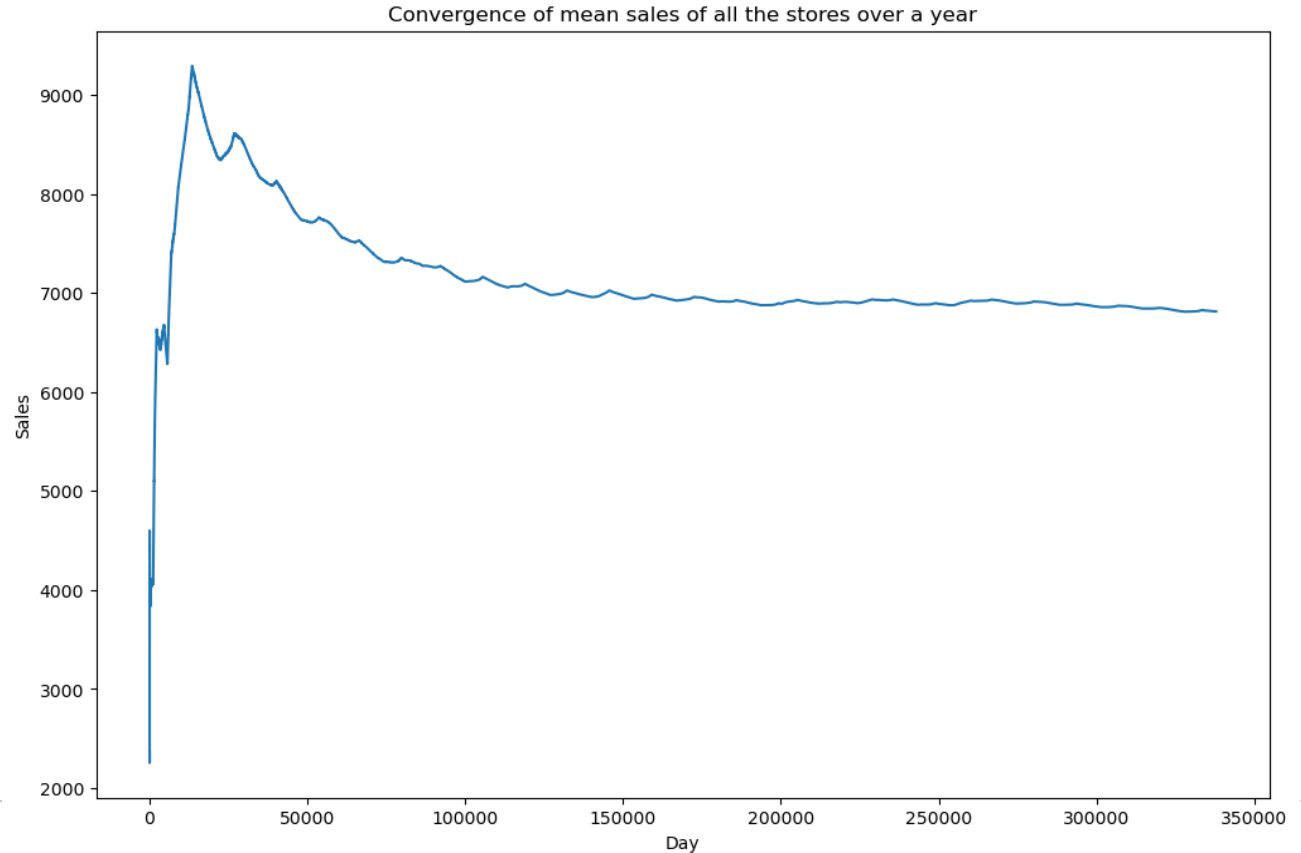
$P(\text{HighSales} \mid \text{Promo}=0): 3.87 \%$

$P(\text{LowSales} \mid \text{Promo}=1): 0.82 \%$

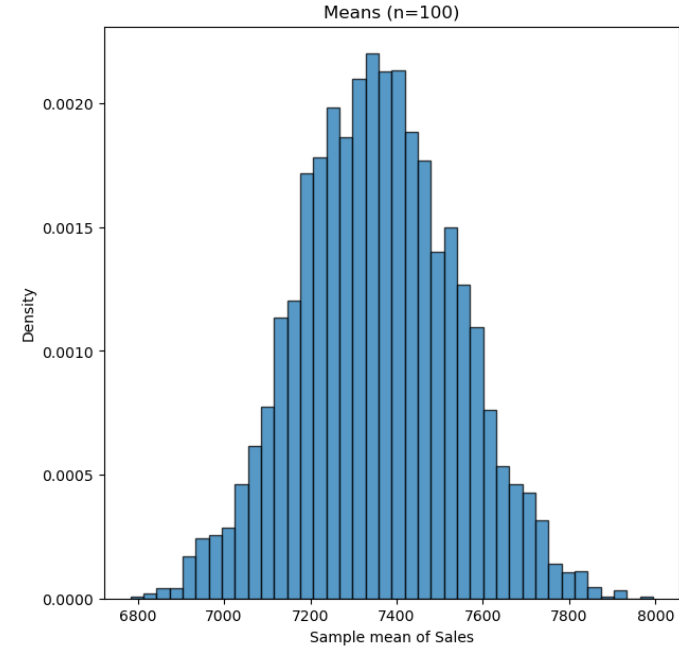
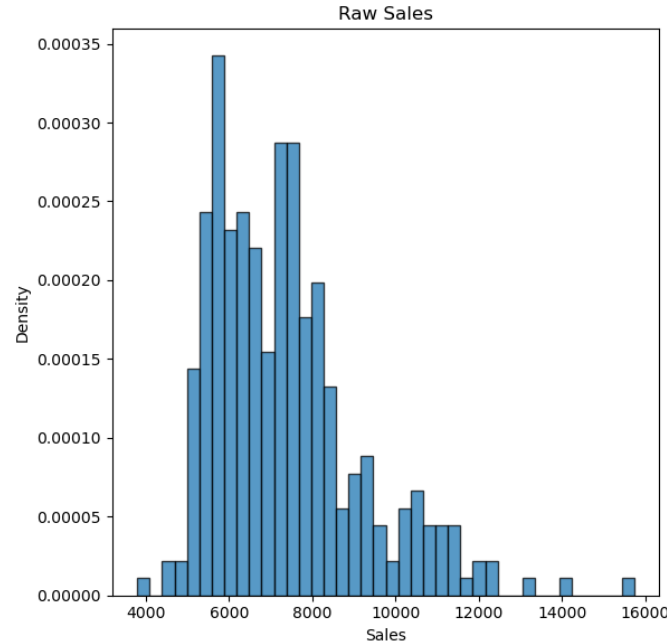
$P(\text{LowSales} \mid \text{Promo}=0): 17.42 \%$

IsWeekend	0	1
Promo		
0	54.97	45.03
1	80.45	19.55

Law of Large Numbers



Central Limit Theorem



Actual vs. Predicted sales

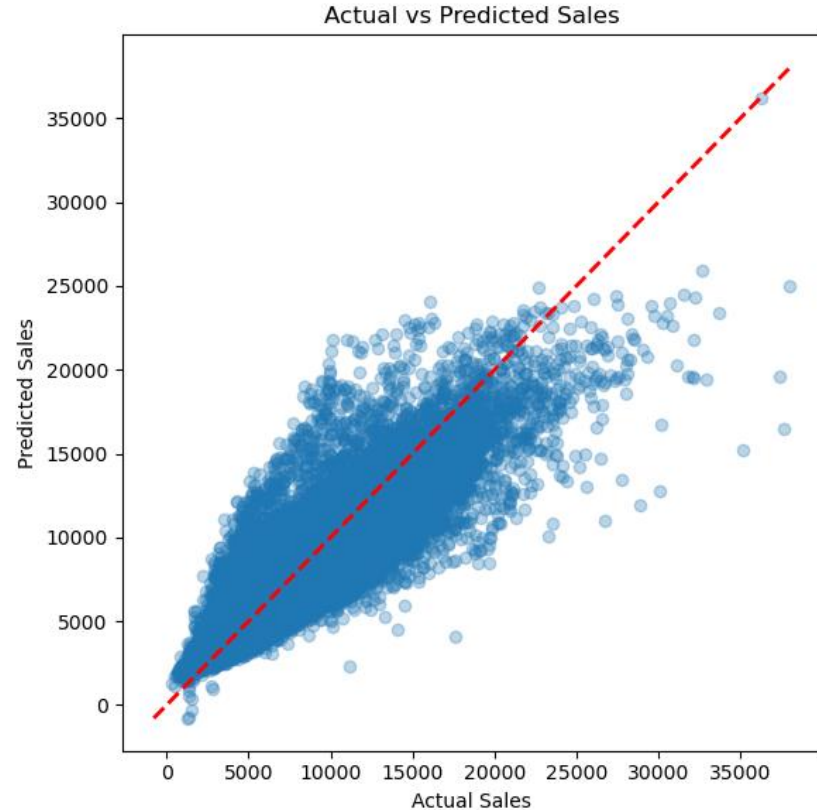
Mean sales: 6814.95
Root mean square error (polynomial-2 linear regression): 1497.67 (21.97%)

Inaccuracy is likely due to:

1. Calendar patterns
2. Sufficiency problems
3. Varying promotions

R^2 : 0.77
MAE: 1076.48

Reference notebooks were searched and algorithms such as Random Forest, XGBoost show accuracy reaching ~90% instead of ~83% here.



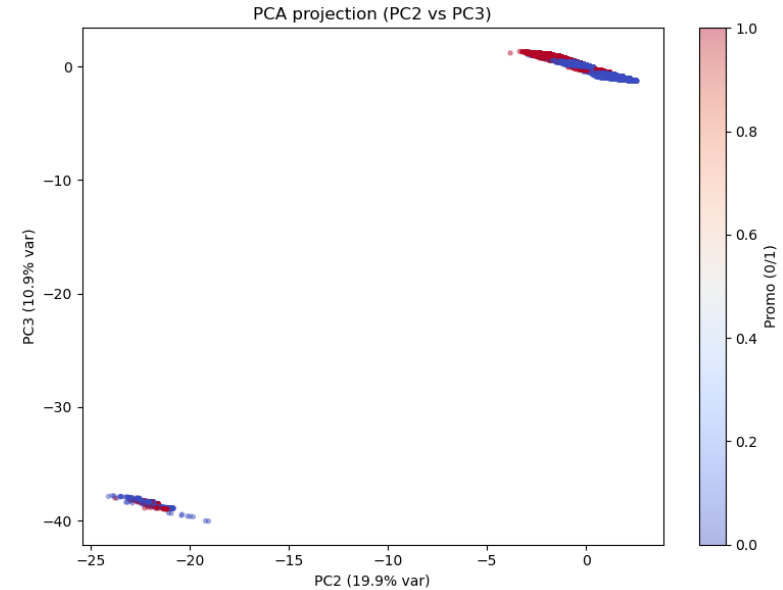
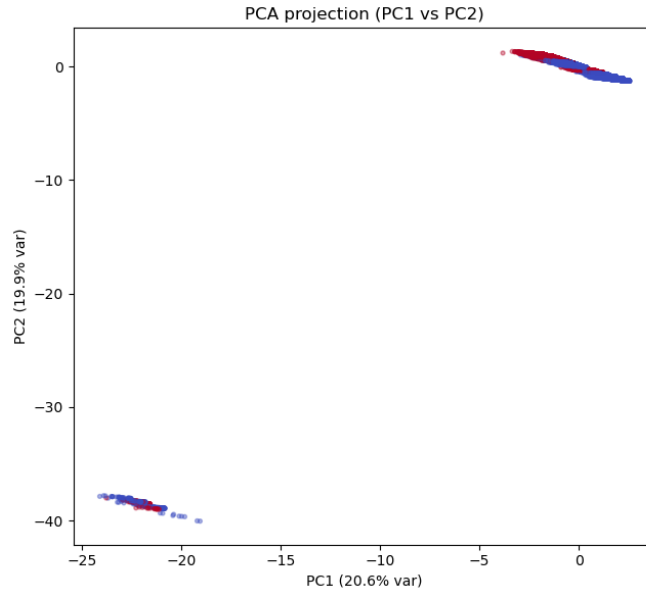
PCA

Explained variance
ratio:

PC1: 20.618%

PC2: 19.853%

PC3: 10.908%



PCA

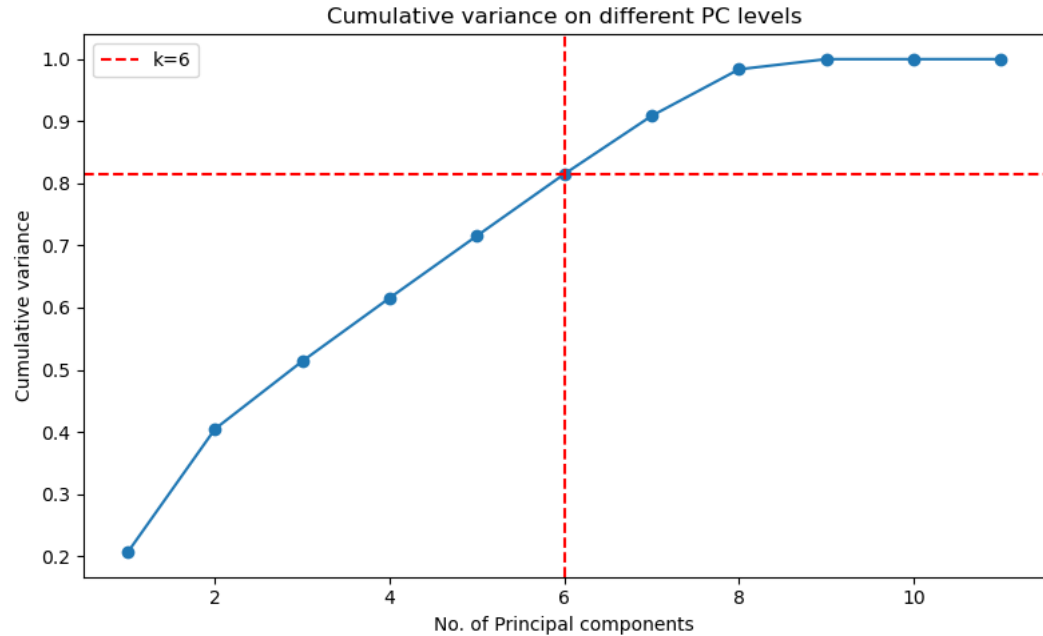
Cumulative variance:

k=2: 40.47%

k=3: 51.38%

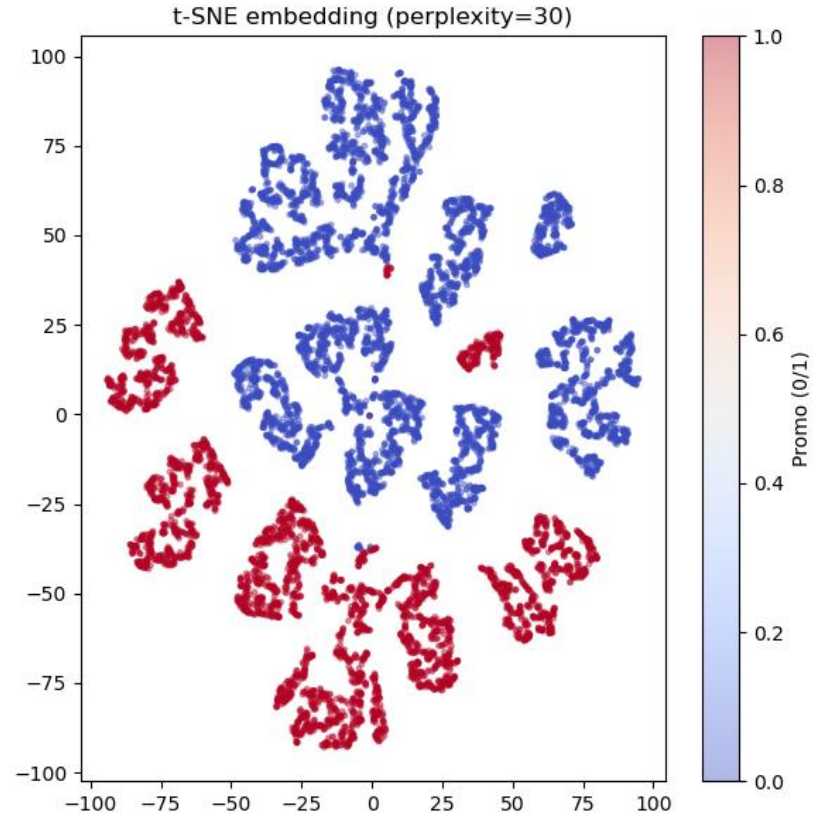
k for 80%: 6

k for 90%: 7



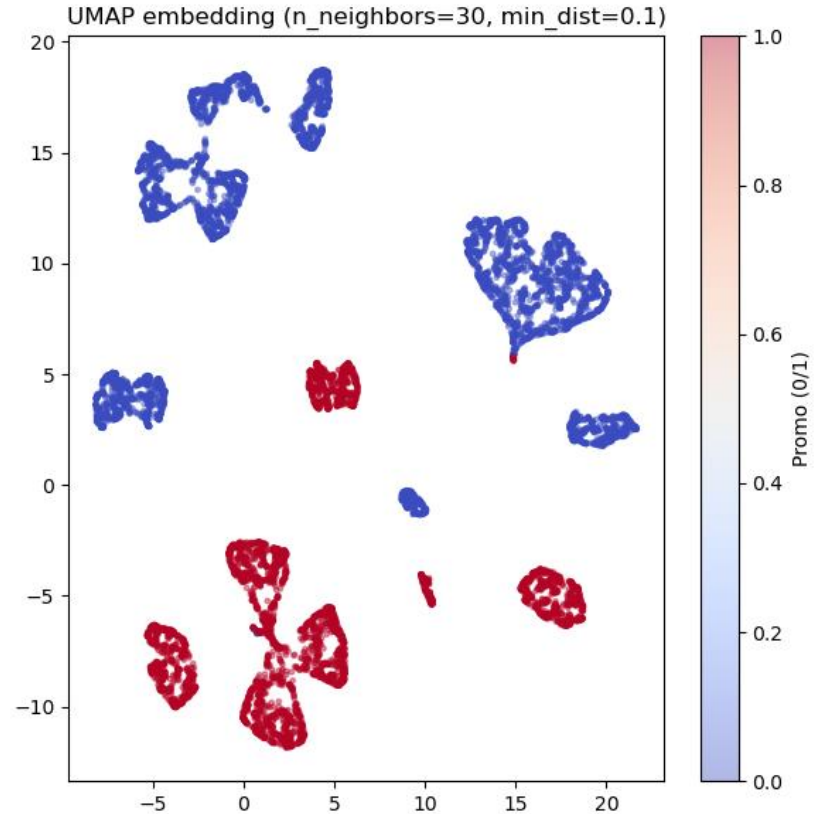
t-SNE

- Since we see that Promo regions are different from non-promo regions, there's a fundamental difference between the two.



UMAP

- We observe the local neighborhoods being preserved.



Chi-square test

Null Hypothesis: The events “IsWeekend” and “Promo” are independent.

Results for the test:

$\chi^2=24192.940$

dof=1

p-value=0

Hypothesis rejected, explanation:

Chances of Promo happening on a weekend:
20%.

Chances of Promo happening on a weekday:
45%.

IsWeekend	0	1
Promo		
0	104492	86444
1	118060	28928

QUESTIONS?